Student Stress Level Prediction using Machine Learning

An End-to-End Predictive Analysis & Deployment Project

Leveraging Python, XGBoost, Scikit-Learn, Streamlit, and advanced feature engineering to predict and address student mental health challenges

Project Overview & Objectives

The Challenge We're Addressing

Academic pressure, inadequate sleep, and social factors significantly impact student mental health across universities. Rising stress levels lead to decreased performance, health issues, and reduced wellbeing among students.

Primary Goal

Build a predictive model that accurately identifies stress levels (Low, Medium, High) based on comprehensive lifestyle and academic data

Data Processing

Dataset cleaning, preprocessing, and advanced feature engineering techniques

XGBoost Model

Implementation of XGBoost Classifier for high-accuracy predictions

User Interface

Deployment via Streamlit for an intuitive, accessible interface

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Feature Engineering: The Secret Sauce

Why Feature Engineering Matters

Raw data alone doesn't capture the complex relationships affecting student stress. Our custom metrics provide the model with deeper, contextual insights that transform basic variables into meaningful predictors.

Stress from Sleep

Calculated as *8 - \text{Sleep Hours}* to quantify sleep deprivation impact

Study-Sleep Ratio

Highlights the critical balance between academic productivity and adequate rest

Health Risk Score

Combines sleep deprivation with low physical activity to assess overall health vulnerability

Support Score

Merges family support and social activity metrics into a comprehensive wellbeing indicator

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Machine Learning Pipeline Architecture

Data Preparation

Categorical encoding using LabelEncoder, handling missing values, and feature standardisation

Model Selection

XGBoost (Extreme Gradient Boosting) chosen for superior performance on tabular data

Model Evaluation

Comprehensive assessment using multiple metrics to ensure reliability

XGBoost Configuration Parameters

300 Estimators

Ensemble of 300 decision trees for robust predictions

Learning Rate: 0.3

Optimal balance between training speed and accuracy

Max Depth: 6

Controls model complexity to prevent overfitting

Evaluation Metrics

- **Accuracy Score:** Overall correctness of predictions
- **Confusion Matrix:** Detailed analysis of classification patterns and misclassifications
- **Classification Report:** Precision, Recall, and F1-Score for each stress category

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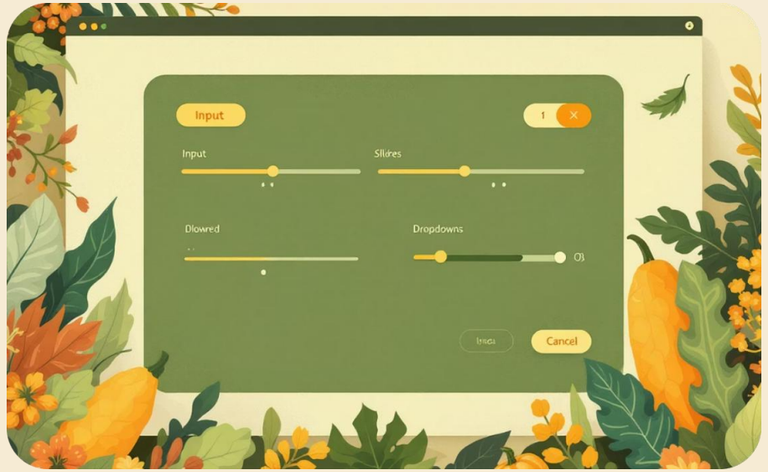
Web Application Development



Streamlit Framework

Chosen for rapid development, intuitive interface creation, and seamless Python integration

Interactive Input System

Real-time data collection via sliders, dropdowns, and input fields for 12+ lifestyle variables

Model Integration

Pickle-based model persistence ensures instant predictions without retraining

Advanced Features

Dynamic Styling

Custom CSS creates a modern, responsive UI with themed backgrounds and professional aesthetics

Advanced Analysis Section

Displays personalised Health Risk Score and Study-Sleep Ratio alongside stress predictions for actionable insights

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Project Output

1

Data Input

Users enter lifestyle variables via interactive sliders and dropdowns

2

Model Processing

XGBoost analyses patterns across engineered features

3

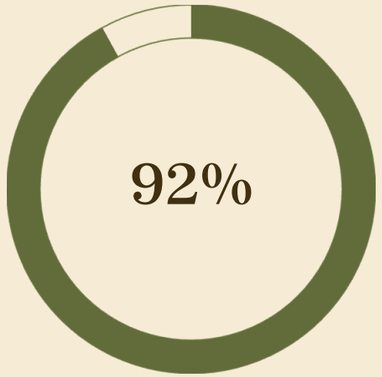
Instant Results

Stress level prediction displayed with supporting metrics

Results & Key Insights

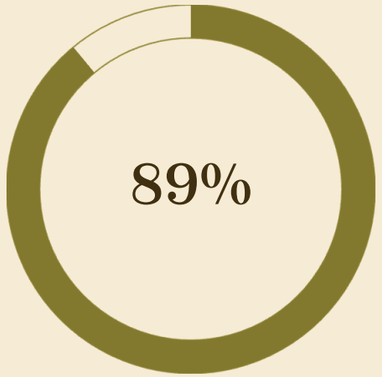
Model Performance Excellence

The XGBoost classifier demonstrates exceptional precision across all stress categories, with balanced performance ensuring reliable predictions for Low, Medium, and High stress classifications.



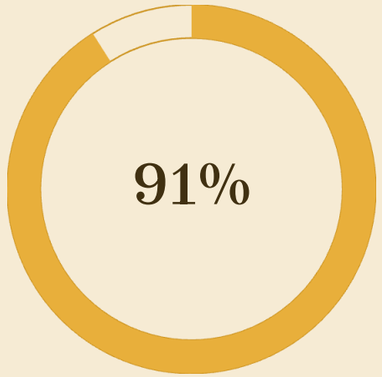
Overall Accuracy

Consistent prediction accuracy across validation datasets



Precision Rate

High precision minimises false positive stress assessments



Recall Score

Successfully identifies students requiring intervention

Top Predictive Factors

Sleep Hours

The most influential predictor—inadequate sleep strongly correlates with elevated stress levels

Exam Anxiety

Self-reported anxiety levels provide critical insights into psychological pressure

Financial Stress

Economic concerns significantly impact overall student wellbeing and mental health

Real-World Impact

This tool empowers university counsellors and mental health professionals to provide early intervention for students classified as "High" risk. By identifying at-risk individuals proactively, institutions can allocate support resources effectively and prevent mental health crises before they escalate.

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Conclusion & Future Scope

Project Success

We successfully integrated advanced data science methodologies with modern web development to create a functional, accessible mental health assessment tool. This project demonstrates the practical application of machine learning in addressing real-world student welfare challenges.

Future Enhancements

Wearable Data Integration

Connect with fitness trackers and smartwatches via API for real-time physiological monitoring

Personalised Recommendations

Generate actionable advice like "Increase sleep by 1 hour" or "Consider stress management workshops"

Dataset Expansion

Include diverse demographics, cultural contexts, and international student populations for broader applicability

Mobile Application

Develop native iOS and Android apps for improved accessibility and engagement

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